Gravity-Torque Momentum Management for a Three-Wheel Reaction-Wheel Inverted Pendulum

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***Abstract*—A reaction wheel that holds a body against a persistent torque integrates that torque, so its speed, not its torque, is what runs out. On a pendulum balancing about a ground pivot the only external torque that can unload the wheels is gravity, which also destabilizes the plant. We show that any equilibrium of the disturbed plant holds the attitude at which gravity balances the disturbance, with no momentum accumulating in the wheels, so designs that regulate the attitude error to zero have no equilibrium to reach. A $2\,\mathrm{mm}$ center-of-mass offset needs under 1° of lean, yet it fells feedback linearization, sliding mode control, backstepping and PID, and an integrator on the attitude error does not help. On a nonlinear three-wheel model with a torque–speed envelope, a digital loop and sensor noise, over 150 simulated plants with random center-of-mass offsets and a $15\,\mathrm{s}$ horizon, these designs balance 33.3–36.0 % of the plants; a gravity-torque unloading augmentation raises them to 82.0–96.7 %; the best is level with a constrained MPC (96.7 %), and none outperforms it or an LQR weighted on stored momentum (99.3 %).**

***Index Terms*—underactuated systems, reaction wheels, inverted pendulum, momentum management, actuator constraints, model predictive control, sliding mode control, backstepping**

I. Introduction

The reaction-wheel pendulum is a standard benchmark for nonlinear control [1], and the Cubli [2], [3] extends it to balancing a cube on one vertex with three orthogonal wheels; comparisons of LQR, feedback linearization, sliding mode control and backstepping on this plant are common [4]. They usually model the actuator as a torque saturation. A wheel that holds a body against a persistent torque, however, *integrates* it, and its speed grows until no reaction torque is left in that direction; wheel-speed excess is the reported failure mode on a Cubli-type rover [5].

Spacecraft handle this with momentum management [6], unloading the wheels through an external torque [7] or constraining wheel momentum directly [8]. For balancing machines, the invariance of the Cubli's vertical momentum is known [3], a ballbot holds a center-of-mass offset by leaning [9], a reaction-wheel bicycle balances under lateral load [10], and reference governors enforce constraints through set-points [11]. What is specific to a body balancing about a ground pivot is that the only unloading torque is the *same first-order gravity torque* that destabilizes it: in orbit unloading costs pointing accuracy, here it costs stability margin.

We ask which designs keep such a pendulum balanced under a persistent disturbance when wheel speed, not torque, is the binding limit, and contribute: (i) an equilibrium condition (Proposition 2): every equilibrium of the disturbed plant holds a gravity-balanced lean, to first order $\|P\boldsymbol{\tau}_d\|/(Mgl)$, so designs regulating $\boldsymbol{\theta}_e \to 0$ have no equilibrium to reach; (ii) a directional capacity bound (Proposition 1; cf. [12]) that corrects the scalar budget by $\|\boldsymbol{n}\|_1$; (iii) a measurement of which limit binds: classical designs fail at $\eta_\Omega \approx 0.83$, where the accelerating envelope no longer covers bearing friction; and (iv) a comparison of five designs, each classical, with integral action and with momentum management, against a constrained MPC, all tuned by one protocol (Secs. IV–VI).

II. Problem Formulation

A. Geometry and states

The testbed is a cube of edge L carrying three identical reaction wheels whose spin axes coincide with the body axes $\boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3$ (Fig. 1). It pivots about one vertex, treated as an ideal spherical joint. Let $R \in \mathrm{SO}(3)$ rotate body coordinates into the inertial frame, $\boldsymbol{\omega} \in \mathbb{R}^3$ be the body angular velocity, and $\boldsymbol{\Omega} \in \mathbb{R}^3$ collect the wheel speeds *relative to the body*.

Let $\boldsymbol{J}$ be the inertia of the whole assembly about the pivot, wheels included as if rigidly attached, $\boldsymbol{J}_w = J_a I_3$ the wheel axial inertias, and $\boldsymbol{J}_b = \boldsymbol{J} - \boldsymbol{J}_w$. The total angular momentum about the pivot, in body coordinates, is

$$\boldsymbol{h} = \boldsymbol{J}\boldsymbol{\omega} + \boldsymbol{J}_w\boldsymbol{\Omega}. \tag{1}$$

B. Equations of motion

Wheel torques are internal, so the only external actions about the pivot are gravity and pivot friction. Writing $\boldsymbol{\tau}_g(R) = M\,\boldsymbol{r}_c \times R^\top \boldsymbol{g}$ for the gravity torque, with $\boldsymbol{r}_c$ the vector from pivot to center of mass,

$$\boldsymbol{J}\dot{\boldsymbol{\omega}} + \boldsymbol{J}_w\dot{\boldsymbol{\Omega}} = \boldsymbol{\tau}_g(R) - \boldsymbol{\omega} \times \boldsymbol{h} - b_p\,\boldsymbol{\omega}, \tag{2}$$

$$\boldsymbol{J}_w(\dot{\boldsymbol{\Omega}} + \dot{\boldsymbol{\omega}}) = \boldsymbol{\tau} - \boldsymbol{\tau}_f(\boldsymbol{\Omega}), \tag{3}$$

where $\boldsymbol{\tau}$ are the wheel torques and $\boldsymbol{\tau}_f$ collects viscous and Coulomb bearing friction. Eliminating $\dot{\boldsymbol{\Omega}}$ gives the form that is integrated,

$$\boldsymbol{J}_b\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_g(R) - \boldsymbol{\omega} \times \boldsymbol{h} - b_p\boldsymbol{\omega} - \boldsymbol{\tau} + \boldsymbol{\tau}_f, \tag{4}$$

$$\dot{\boldsymbol{\Omega}} = \boldsymbol{J}_w^{-1}(\boldsymbol{\tau} - \boldsymbol{\tau}_f) - \dot{\boldsymbol{\omega}}, \tag{5}$$

$$\dot{R} = R\,[\boldsymbol{\omega}]_\times\,. \tag{6}$$

No small-angle approximation is made; the attitude is propagated as a unit quaternion. Setting $\boldsymbol{\tau} = 0$ and all friction to zero renders the system conservative, which is the first of the verification tests reported in Sec. V.

C. Actuator and implementation model

Three limits act on the commanded torque, and they are not interchangeable. Besides the peak torque $|\tau_i| \le \tau_{\max}$ and a slew-rate limit, a brushless drive obeys a torque–speed envelope: the torque available in the *accelerating* direction decays as the wheel approaches $\Omega_{\max}$,

$$\tau_i^{\text{avail}}(\Omega_i) = \tau_{\max} \max\left(0,\, 1 - \frac{|\Omega_i|}{\Omega_{\max}}\right), \tag{7}$$

while braking torque remains fully available. The controller runs at $f_s = 500\,\text{Hz}$ with zero-order hold and no computation delay; the wheel speed is reconstructed from a quantized encoder and the rates come from a gyro with white noise and a per-run bias. All results below are obtained in this digital loop, not in continuous time with exact state feedback.

D. Standing assumptions and control objective

(A1) The vertex contact is an ideal spherical joint: it transmits no torque about the pivot other than viscous friction $-b_p\boldsymbol{\omega}$ with $b_p \ge 0$, and the contact does not slip.

(A2) Body and wheels are rigid; each wheel is balanced and axisymmetric about its spin axis, and the three spin axes coincide with the body axes.

(A3) The delivered torque respects (7), the peak torque and the slew limit.

(A4) Attitude and body rates are available at f_s with white Gaussian noise and a constant per-run gyro bias, and the controller knows the plant parameters only up to a bounded multiplicative uncertainty and an unknown center-of-mass offset. No estimator is modelled and no constant attitude-estimate bias is included.

Definition 1 (Constrained balancing)**.** Given (A1)–(A4), design $\boldsymbol{\tau}(\cdot)$ such that, from initial conditions in a set $\mathcal{S}$, attitude and body rate converge to an admissible equilibrium ($|\Omega_i| \le \Omega_{\max}$; only $\eta_\Omega \le 0.831$ is attainable, Sec. IV-G) on the level set of the vertical momentum $\boldsymbol{e}_z^\top R\boldsymbol{h}$ fixed by the initial condition (slowly drifting if $b_p > 0$), *and* $|\Omega_i(t)| \le \Omega_{\max}$ for all $t \ge 0$ and all i.

Convergence is asked of a level set, not of an isolated point, because that momentum is conserved (Lemma 1) and cannot be regulated (Corollary 1). The simulations below evaluate finite-horizon recovery; they do not certify this property. Definition 1 scopes this paper to local balancing and momentum management about the corner equilibrium; global swing-up and its handover are outside that scope and are assumed to be discharged by a separate transient controller that delivers the state into $\mathcal{S}$. The wheel-speed constraint is not an implementation detail appended to a stability requirement: Sec. III shows it is the binding one.

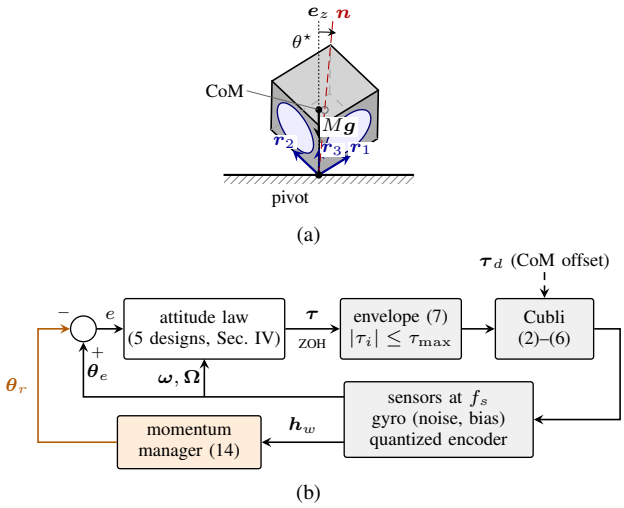


Fig. 1. (a) Idealised side view of the testbed (not to scale): a cube on one vertex (ideal pivot, (A1)), wheel axes $\boldsymbol{r}_1$–$\boldsymbol{r}_3$ along the edges leaving the pivot, and the balancing axis $\boldsymbol{n}$. At the equilibrium of Proposition 2 under a center-of-mass offset, the true center of mass (filled) is above the pivot and the geometric center (open) leans with $\boldsymbol{n}$ by $\theta^\star$ (exaggerated; at 2 mm, $\theta^\star = 0.64\,^\circ$). (b) Simulated loop shared by all designs; the momentum manager (14) only shifts the attitude reference, and the MPC replaces law and manager.

III. Main Results: The Momentum Budget

Lemma 1 (Momentum invariant and its drift)**.** *Consider* (2)–(6) *with* $b_p \ge 0$ *and let* $\boldsymbol{e}_z$ *be the vertical. Then, for* any *wheel torque* $\boldsymbol{\tau}(\cdot)$*,*

$$\frac{d}{dt}\left(\boldsymbol{e}_z^\top R\boldsymbol{h}\right) = -\,b_p\,\boldsymbol{e}_z^\top R\boldsymbol{\omega}. \tag{8}$$

For an ideal pivot ($b_p = 0$) *the quantity* $\boldsymbol{e}_z^\top R\boldsymbol{h}$ *is therefore constant along every trajectory, and for* $b_p > 0$ *its drift is bounded by* $b_p \int_0^t \|\boldsymbol{\omega}\|\, ds$*.*

Proof. $\frac{d}{dt}(R\boldsymbol{h}) = R(\dot{\boldsymbol{h}} + \boldsymbol{\omega} \times \boldsymbol{h}) = R\,\boldsymbol{\tau}_g - b_p R\boldsymbol{\omega}$ by (2). Since $\boldsymbol{\tau}_g = M\boldsymbol{r}_c^I \times \boldsymbol{g}$ with $\boldsymbol{g} \parallel \boldsymbol{e}_z$, we have $\boldsymbol{e}_z^\top R\boldsymbol{\tau}_g = 0$, which leaves (8). The wheel torques never appear: they are internal and cancel in (2). □

The invariance is established in [3]; on the simulated plant ($b_p = 1 \times 10^{-3}\,\text{N m s rad}^{-1}$) the drift stays below $1 \times 10^{-6}\,\text{N m s}$ over 20 s, orders of magnitude less than the momentum Sec. IV-G moves.

The wheels are a *store*, not a sink. The storable set is the *box* $\mathcal{H} = \{\boldsymbol{h}_w : \|\boldsymbol{h}_w\|_\infty \le h_{\max}\}$ with $h_{\max} = J_a\Omega_{\max} = 0.150\,\text{N m s}$ per wheel, so saturation is measured componentwise by $\eta_\Omega := \max_i |\Omega_i|/\Omega_{\max} = \|\boldsymbol{h}_w\|_\infty / h_{\max}$ and (A3) reads $\eta_\Omega \le 1$. The Euclidean norm is the wrong index: $\|\boldsymbol{h}_w\|_2$ can reach $\sqrt{3}\,h_{\max}$ inside $\mathcal{H}$.

Proposition 1 (Directional capacity)**.** *For any unit* $\boldsymbol{n}$*,* $\max_{\boldsymbol{h}_w \in \mathcal{H}} \boldsymbol{n}^\top \boldsymbol{h}_w = h_{\max}\|\boldsymbol{n}\|_1$*, attained at the vertex* $\Omega_i = \Omega_{\max}\,\text{sign}(n_i)$*. For the corner equilibrium of a cube,* $\boldsymbol{n} = (1,1,1)/\sqrt{3}$ *and the capacity along the balancing axis is* $\sqrt{3}\,h_{\max} = 0.260\,\text{N m s}$*.*

Proof. $\boldsymbol{n}^\top \boldsymbol{h}_w = J_a \boldsymbol{n}^\top \boldsymbol{\Omega}$ is linear, so its maximum over a box is attained at a vertex; evaluating gives $J_a\Omega_{\max}\sum_i |n_i|$. □

Corollary 1 (Loss of stabilizability)**.** *Let $\boldsymbol{n}$ be the balancing axis in body coordinates and linearize* (4)–(6) *about the corner equilibrium with no stored wheel momentum (or momentum along $\boldsymbol{n}$), with states $z = (\boldsymbol{\theta}_e, \boldsymbol{\omega}, \boldsymbol{h}_w)$, $\boldsymbol{h}_w = \boldsymbol{J}_w\boldsymbol{\Omega}$, yaw included. Then for* every *$b_p \geq 0$ the row vector*

$$v(b_p) = \begin{bmatrix} b_p\,\boldsymbol{n}^\top & (\boldsymbol{J}\boldsymbol{n})^\top & \boldsymbol{n}^\top \end{bmatrix} \tag{9}$$

satisfies $v(b_p)A = 0$ and $v(b_p)B = 0$. Hence $\operatorname{rank}[A \;\; B] = n_x - 1$*, the uncontrollable eigenvalue is $\lambda_{uc} = 0$ exactly, and (A, B) is* not stabilizable.

Proof. $vB = 0$ because wheel torque enters $\dot{\boldsymbol{\omega}}$ and $\dot{\boldsymbol{h}}_w$ with equal and opposite weight, which $(\boldsymbol{J}\boldsymbol{n}, \boldsymbol{n})$ annihilates. $vA = 0$ states that $\boldsymbol{n}^\top(\boldsymbol{J}\boldsymbol{\omega} + \boldsymbol{h}_w) + b_p\,\boldsymbol{n}^\top\boldsymbol{\theta}_e$ is conserved to first order: the friction torque $-b_p\boldsymbol{\omega}$ integrates to $-b_p\boldsymbol{\theta}_e$, so the momentum it removes is exactly the b_p term. Thus v is a left eigenvector of A at $\lambda = 0$ and $vB = 0$ makes the PBH test fail there; verified numerically to 7×10^{-16} over $b_p \in [0, 1]$. With transverse stored momentum $\boldsymbol{h}_0$ the annihilator gains the term $(\boldsymbol{h}_0 \times \boldsymbol{n})^\top$ in the attitude block and the rank deficiency persists. □

As noted in [2], pivot friction does *not* restore stabilizability (with yaw free, $b_p > 0$ dissipates vertical momentum only through net yaw rotation, unused here). An LQR posed on the full linearization has no stabilizing Riccati solution; every linear design here is synthesised on the controllable subspace from a Kalman decomposition.

IV. Control design

Six designs act on the full nonlinear plant and see the same actuator model and the same measurements: PID, LQR, feedback linearization with an LQR outer loop (FBL+LQR), sliding mode control (SMC), backstepping and constrained model predictive control (MPC).

A. Shared cancellation term and notation

The three nonlinear laws share the cancellation term

$$\Phi(x) = \boldsymbol{\tau}_g(R) - \boldsymbol{\omega} \times \boldsymbol{h} - b_p\boldsymbol{\omega} + \boldsymbol{\tau}_f(\boldsymbol{\Omega}), \tag{10}$$

so that (4) reads $\boldsymbol{J}_b\dot{\boldsymbol{\omega}} = \Phi(x) - \boldsymbol{\tau}$ and the designs differ in feedback shaping rather than in how much physics each is allowed to cancel. Let $\boldsymbol{\theta}_e$ denote the attitude error and $\boldsymbol{\theta}_r$ a reference tilt defined in Sec. IV-G, with $e = \boldsymbol{\theta}_e - \boldsymbol{\theta}_r$.

B. Linear-class baselines: PID and LQR

PID acts on attitude with rate damping and integrator clamping; it is the declared linear-class baseline. The LQR is a Riccati design on the controllable subspace of the linearization about the corner equilibrium, with states $(\boldsymbol{\theta}_e, \boldsymbol{\omega}, \boldsymbol{h}_w)$ and weights $Q = \operatorname{diag}(q_\theta I, q_\omega I, q_h I)$, $R = rI$. The weight q_h acts directly on stored momentum; this is what makes the LQR implicitly budget-aware.

C. Feedback linearization with an LQR outer loop

Cancelling $\Phi(x)$ leaves $\dot{\boldsymbol{\omega}} = v$ at any attitude for the nominal plant, and v must still be chosen. The error kinematics are only first order: with $\boldsymbol{\theta}_e = 2\operatorname{vec}(q_e)$ and $q_e = (\eta, \boldsymbol{\epsilon})$ one has $\dot{\boldsymbol{\theta}}_e = \eta\boldsymbol{\omega} + \frac{1}{2}\boldsymbol{\theta}_e \times \boldsymbol{\omega}$, so $\dot{\boldsymbol{\theta}}_e = \boldsymbol{\omega}$ — and the double-integrator structure — holds near the equilibrium (a $2.6\,\%$ relative error at $3°$, $39\,\%$ at the $45°$ fall threshold). The outer loop is the LQR for that double integrator,

$$v = -(k_1 e + k_2\boldsymbol{\omega}), \quad \boldsymbol{\tau} = \Phi(x) + \boldsymbol{J}_b(k_1 e + k_2\boldsymbol{\omega}), \tag{11}$$

with (k_1, k_2) from the Riccati equation of $\left(\left[\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right], \left[\begin{smallmatrix}0\\ 1\end{smallmatrix}\right]\right)$ and weights (q_θ, q_ω, r). The wheel speeds are the zero dynamics of the design and are left unregulated.

D. Sliding mode control

With sliding variable $s = \boldsymbol{\omega} + \Lambda e$ and boundary layer ϕ,

$$\boldsymbol{\tau} = \Phi(x) + \boldsymbol{J}_b(\Lambda\boldsymbol{\omega} + K\,\mathrm{sat}(s/\phi)). \tag{12}$$

A boundary layer replaces sign, whose chatter on a sampled loop would inject momentum.

E. Backstepping

With $z_1 = e$, $\boldsymbol{\omega}_d = -k_1 z_1$, $z_2 = \boldsymbol{\omega} - \boldsymbol{\omega}_d$ and $\Gamma = \gamma I \succ 0$ in $\mathrm{N\,m\,rad^{-1}}$ so both terms of $V = \frac{1}{2}z_1^\top\Gamma z_1 + \frac{1}{2}z_2^\top\boldsymbol{J}_b z_2$ are energies,

$$\boldsymbol{\tau} = \Phi(x) + k_1\boldsymbol{J}_b\boldsymbol{\omega} + K_2 z_2 + \Gamma z_1 \tag{13}$$

gives, for constant $\boldsymbol{\theta}_r$, $\dot{V} = -k_1 z_1^\top\Gamma z_1 - z_2^\top K_2 z_2 + \Delta$ with $\Delta = \mathcal{O}(\|z\|^3)$ vanishing at $\eta = 1$: stability is local.

F. Constrained model predictive control

MPC solves, nominally every $20\,\mathrm{ms}$ (about $47\,\mathrm{Hz}$ effective, a floating-point scheduling artefact of our code) over a $1\,\mathrm{s}$ horizon, a quadratic program in z with $|\tau_i| \leq \tau_{\max}$ and $|h_{w,i}| \leq h_{\max}$ imposed on every predicted step and a terminal cost from the discrete Riccati equation on the controllable subspace. It is the only design here that is *told* about the momentum limit, although in our runs that bound never becomes active.

Every design except MPC also exists in an *integral* variant (for the PID, which already integrates, it coincides with the baseline), $\boldsymbol{\tau}\mathrel{+}= K_i\int e\,dt$ with clamping, and in a *momentum-managed* variant defined next.

G. Momentum management

Lemma 1 also says how to unload the wheels: since the wheels cannot remove momentum, an external torque must, and the only one available is gravity. To first order the gravity torque is $\boldsymbol{\tau}_g \simeq Mgl(I - \boldsymbol{n}\boldsymbol{n}^\top)\boldsymbol{\theta}_e$, so holding the tilt

$$\boldsymbol{\theta}_r = -\frac{k_h}{Mgl}\,(I - \boldsymbol{n}\boldsymbol{n}^\top)\,\boldsymbol{h}_w \tag{14}$$

produces, to first order, the torque that decelerates the wheels. The commanded lean is clamped at $3°$. The projector removes the component along $\boldsymbol{n}$, precisely the one Lemma 1 forbids

removing for $b_p = 0$; $k_h = 0$ recovers the classical designs, so the augmentation is measured rather than assumed.

Proposition 2 (Equilibrium lean and saturation time)**.** *Let a torque $\boldsymbol{\tau}_d(R)$ act about the pivot. (i)* (exact) *At any equilibrium $(R^\star, \boldsymbol{\omega} = 0, \boldsymbol{\Omega}^\star)$ of* (4)–(6)*, $\boldsymbol{\tau}_g(R^\star) + \boldsymbol{\tau}_d(R^\star) = 0$ and $\boldsymbol{\tau} = \boldsymbol{\tau}_f(\boldsymbol{\Omega}^\star)$: gravity carries the disturbance, the motors only cover bearing friction, and no momentum accumulates. (ii)* (first order, $\boldsymbol{n}^\top \boldsymbol{\tau}_d = 0$, small $\boldsymbol{\tau}_d$) *The tilt part of the equilibrium attitude error satisfies*

$$P\boldsymbol{\theta}^\star = -\frac{1}{Mgl}\, P\,\boldsymbol{\tau}_d, \qquad P = I - \boldsymbol{n}\boldsymbol{n}^\top, \tag{15}$$

(iii) (first order, $\boldsymbol{n}^\top \boldsymbol{\tau}_d = 0$) *A controller that regulates $\boldsymbol{\theta}_e \to 0$ admits no equilibrium under $P\boldsymbol{\tau}_d \neq 0$: its stored momentum grows as $\dot{\boldsymbol{h}}_w \simeq P\boldsymbol{\tau}_d$ and, from zero momentum, reaches a saturation index $\eta^\star$ after*

$$t_{sat} \;\simeq\; \frac{\eta^\star h_{\max}}{\|P\boldsymbol{\tau}_d\|_\infty}. \tag{16}$$

Proof. (i) With $\boldsymbol{\omega} \equiv 0$, (2) gives $\boldsymbol{J}_w\dot{\boldsymbol{\Omega}} = \boldsymbol{\tau}_g + \boldsymbol{\tau}_d$, which must vanish, and then (3) gives $\boldsymbol{\tau} = \boldsymbol{\tau}_f$. (ii) Use $\boldsymbol{\tau}_g \simeq MglP\boldsymbol{\theta}_e$; the equation is solvable only for $\boldsymbol{\tau}_d \perp \boldsymbol{n}$ (gravity has no torque about $\boldsymbol{n}$). (iii) If $\boldsymbol{\theta}_e \equiv 0$ then $\boldsymbol{\tau}_g = 0$ and, by total momentum, $\dot{\boldsymbol{h}}_w = \boldsymbol{\tau}_d - \boldsymbol{J}\dot{\boldsymbol{\omega}} \simeq P\boldsymbol{\tau}_d$. □

For an offset d along a body axis, $\boldsymbol{\tau}_d = -Mgd\,(\boldsymbol{e}_i \times \boldsymbol{n})$, so $\|P\boldsymbol{\tau}_d\|_2 = Mgd\sqrt{2/3}$ and $\|P\boldsymbol{\tau}_d\|_\infty = Mgd/\sqrt{3}$: the component of the offset along $\boldsymbol{n}$ tilts nothing. The required lean is therefore $\theta^\star \simeq \sqrt{2/3}\,d/l = 0.64^\circ$ at $d = 2\,\mathrm{mm}$ (exactly, $\operatorname{atan2}(|d|\|P\boldsymbol{e}_i\|, l + d\,n_i)$); (16) takes the *infinity* norm because the budget index is component-wise. Part (i) adds a necessary condition: the motor torque on a wheel at the equilibrium speed, $\tau_f(\Omega^\star)$, is accelerating, so by (7) $b_w|\Omega_i^\star| + \tau_c \le \tau_{\max}(1 - |\Omega_i^\star|/\Omega_{\max})$, i.e. $|\Omega_i^\star| \le 831\,\mathrm{rad\,s^{-1}}$ ($\eta_\Omega \le 0.831$) on the nominal plant.

H. Tuning protocol

Every controller is tuned by the same procedure, so that no baseline is handicapped by its gains: the same cost

$$J_c = \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \begin{cases} w_e\!\int\!\|\boldsymbol{\theta}_e\|\,dt + w_u\!\int\!\|\boldsymbol{\tau}\|^2 dt, & \text{balanced,} \\ \rho\,(1 - t_{\text{fall}}/T), & \text{fallen,} \end{cases} \tag{17}$$

with $w_e = 1$, $w_u = 20$, $\rho = 10^3$: a fallen run is scored by survival time alone, so truncated error cannot reward failing early. The branches are not strictly ordered: a fall in the last $2\,\%$ of the horizon can score below a survival. A survival-first (lexicographic) re-tuning leaves the MPC gains unchanged and finds no gain set with which a momentum-blind design survives the asymmetric scenario (Sec. VI). The protocol also fixes the same scenario set $\mathcal{S}$ (small and large transients, an asymmetric plant, a pre-loaded wheel), the same Nelder–Mead optimiser on log-gains and the same iteration cap, never reached (42–102 cost evaluations per design). All gains are listed in the supplementary material.

V. Simulation Study

The testbed is a declared design point, not identified hardware: a $15\,\mathrm{cm}$ cube, $M = 1.41\,\mathrm{kg}$, lever arm $l = 14.6\,\mathrm{cm}$, wheel inertia $J_a = 1.50 \times 10^{-4}\,\mathrm{kg\,m^2}$, $\tau_{\max} = 0.50\,\mathrm{N\,m}$ and $\Omega_{\max} = 1000\,\mathrm{rad\,s^{-1}}$, giving $h_{\max} = 0.150\,\mathrm{N\,m\,s}$ per wheel; the full parameter table is in the supplementary material. The model is first checked against properties it must have: energy conserved to 8.9×10^{-13} over $5\,\mathrm{s}$ without friction or torque, the invariant drifting less than $1.1 \times 10^{-11}\,\mathrm{N\,m\,s}$ under arbitrary wheel torques, and the linearization matching a central-difference Jacobian to 7.5×10^{-9}; the step is $1\,\mathrm{ms}$ ($0.67\,\%$ change over $dt \in [0.2, 2]\,\mathrm{ms}$).

The studies sweep the center-of-mass asymmetry, the wheel-speed limit, the peak torque and the pivot friction; map recovery over initial tilt and pre-stored momentum in both storage directions; and run 150 plants with $\pm 20\,\%$ on geometry, mass and dissipation parameters (actuator limits and sensor models fixed) over $15\,\mathrm{s}$. Every controller is designed on the nominal symmetric model; perturbed parameters enter only the simulated plant. Gains come from Sec. IV-H, tuned at $1.5\,\mathrm{mm}$ of offset over $30\,\mathrm{s}$, an operating point that differs from the Monte Carlo and appears in Fig. 3 only as one column.

VI. Results

From a 3° tilt on the nominal plant every controller balances (0.06–0.08 s median time to stay within 2°): this usual regime does not discriminate.

A. A 2 mm asymmetry, and why it is fatal

Give the plant a $2\,\mathrm{mm}$ center-of-mass offset and the classical designs hold attitude perfectly until they lose the cube (Fig. 2). Both halves of Proposition 2 are confirmed numerically. First, the disturbance is small: the equilibrium lean is $\theta^\star \simeq \sqrt{2/3}\,d/l = 0.64^\circ$, and with momentum management the measured steady tilt tracks it to $2\,\%$ over $d \in [1, 3]\,\mathrm{mm}$ and the wheels stop charging. They stop at a standing index that a proportional law necessarily leaves and that grows linearly with d: 0.34, 0.24, 0.31, 0.20 and 0.17 per mm for FBL+LQR, SMC, backstepping, PID and LQR (noise-free $60\,\mathrm{s}$ runs, nominal plant), within $3\,\%$ ($10\,\%$ with sensor noise) of $Mgd/(\sqrt{3}\,k_h h_{\max})$ from (14) and (15) for the four managed designs (FBL+LQR, SMC, backstepping, PID); FBL+LQR falls from $2.5\,\mathrm{mm}$, where that index exceeds 0.831. Second, the classical designs regulate $\boldsymbol{\theta}_e \to 0$, which is not an equilibrium of the disturbed plant, so they must supply $P\boldsymbol{\tau}_d$ forever: their stored momentum ramps linearly and (16) predicts $t_{sat} \propto 1/d$. We measure $t_{sat}d = 16.5\,\mathrm{s\,mm}$ against $15.6\,\mathrm{s\,mm}$ from (16).

B. Integral action on attitude does not rescue them

A persistent disturbance is textbook integral territory, so a comparison in which the nonlinear laws have no integrator would be rigged. Every design therefore has an integral variant, tuned by the same protocol. It does not help: over 150 perturbed plants the balanced-run rate moves from 36.0 to 36.0 % for FBL+LQR, 36.0 to 36.0 % for SMC and 34.7 to

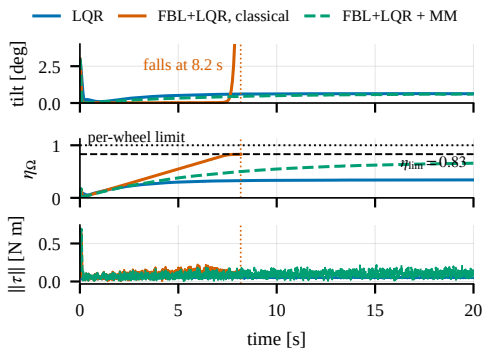


Fig. 2. Corner balancing with a 2 mm center-of-mass offset. The classical FBL+LQR design (orange) tracks attitude until the torque–speed envelope exhausts its accelerating authority, then falls; the same design with momentum management (green, dashed) holds a small tilt that bleeds the wheels through gravity. Middle: component-wise budget index $\eta_\Omega = \max_i |\Omega_i|/\Omega_{\max}$: per-wheel limit dotted, friction–envelope limit $\eta_{\lim}$ dashed. Tilt axis clipped at 4°.

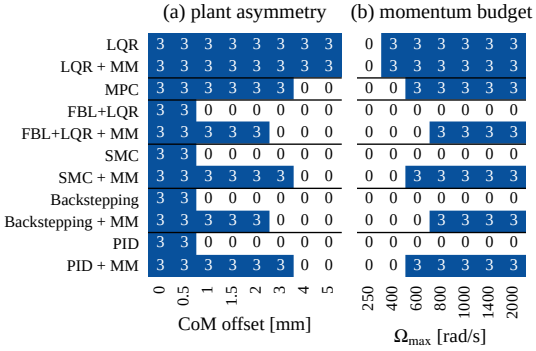


Fig. 3. Runs that balance, out of 3 (sensor-noise seeds, 3° initial tilt, 20 s), against (a) the offset at $\Omega_{\max} = 1000\,\mathrm{rad\,s^{-1}}$ and (b) $\Omega_{\max}$ at 2 mm offset, without and with momentum management (MM).

35.3 % for backstepping, and for the LQR it *falls*, from 99.3 to 87.3 %. The PID already integrates.

The reason is exact: an unsaturated integrator on $\boldsymbol{\theta}_e$ makes $\boldsymbol{\theta}_e = 0$ the only candidate rest point, which Proposition 2(i) excludes when $P\boldsymbol{\tau}_d \neq 0$, so momentum ramps at $P\boldsymbol{\tau}_d$ whatever the gains. Integral action on stored momentum is a different design, not tested here.

In these runs the classical designs never reach the wheel-speed limit. They fail at $\eta_\Omega \approx 0.83$, the terminal speed at which the accelerating envelope (7) only just covers bearing friction ($\eta_\Omega = 0.831$, Sec. IV-G); a model with only a hard speed clamp would have predicted a longer survival. Fig. 3 sweeps the asymmetry and the momentum budget. Sweeping $\Omega_{\max}$, the LQR balances down to 400 $\mathrm{rad\,s^{-1}}$ while no classical nonlinear design balances at *any* budget in the swept range: their failure is not having too little momentum but never releasing what they store. With the augmentation they need 600–800 $\mathrm{rad\,s^{-1}}$.

C. The signature of the invariant

Fig. 4 maps recovery over initial tilt and pre-stored wheel momentum (1 mm offset, 25 s), separating momentum stored perpendicular to the balancing axis, which gravity can dump,

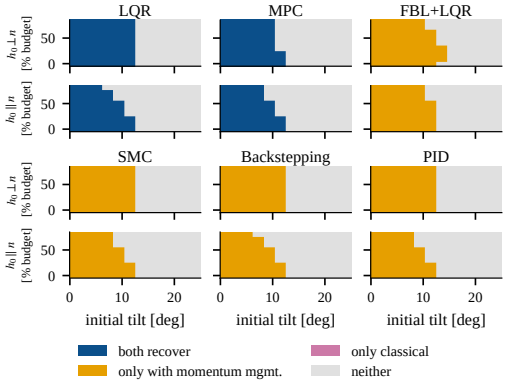


Fig. 4. Empirical recovery map over initial tilt and pre-stored wheel momentum h_0 (1 mm offset, 25 s, one seed per cell; not a certified region of attraction): momentum stored perpendicular to $\boldsymbol{n}$ (dumpable) above momentum along $\boldsymbol{n}$ (the invariant of Lemma 1). Orange marks states only the momentum-managed variant recovers: a band that does not shrink with dumpable momentum and narrows as momentum along the invariant grows. MPC has one variant (blue or grey). No cell is pink: on the sampled grid the augmentation never costs a state the classical design already recovered.

from momentum along the axis, which by Lemma 1 it cannot. Classical FBL+LQR, SMC, backstepping and PID recover 0 % of the sampled grid: at this offset they fall within the horizon from any initial state.

The invariant shows itself in how the augmented recovery depends on what is already stored. With no stored momentum the two directions are indistinguishable (50 against 50 %), as they must be. Pooled over the five managed designs, as the pre-stored momentum grows the dumpable direction is unaffected (49 % beyond 60 % of the budget) while recovery along the invariant falls to 34 %.

D. Perturbed plants, and a constrained benchmark

Table I reports 150 plants with ±20 % on geometry, mass and dissipation parameters and a center-of-mass offset drawn per plant from $\mathcal{N}(1, 1.5^2)$ mm clipped at 3σ (either sign). Episodes last 15 s, so survival is horizon-censored: every momentum-blind survivor has $|d| \leq 1.13$ mm, and managed designs near their standing-momentum limit can also fall later. With the inertias recomputed from the drawn geometry, $h_{\max}$ varies by $-55\,\%/+40\,\%$ even though $\Omega_{\max}$ is fixed.

The three momentum-blind designs balance 34.7–36.0 % of plants. The LQR and the MPC weight stored momentum and the PID integrates, so none of them is momentum-blind. With momentum management, SMC reaches 94.0 %, PID 96.7 %, backstepping 85.3 % and FBL+LQR 82.0 %. The MPC, with no augmentation, reaches 96.7 %. The LQR reaches 99.3 % on its own; adding the augmentation takes it to 100.0 %, a difference that is *not* significant (McNemar $p = 1$); its tuned $k_h = 0.002$ makes the two variants nearly the same controller. With the survival-first re-tuning of Sec. IV-H the classical rates stay at 31.3–35.3 %, MPC at 96.7 %, the LQR moves to 96.7 % (McNemar $p = 0.12$), and momentum management reaches 96.0, 98.0, 96.0 and 98.7 % for FBL+LQR, backstepping, SMC and PID (the first two up from 82.0 and 85.3 %, $p < 10^{-4}$, McNemar; one optimizer run per design): the gain

TABLE I
NOMINAL AND MONTE-CARLO METRICS, WITH WILSON 95 % INTERVALS; MONTE-CARLO RATES DEPEND ON THE OFFSET DISTRIBUTION. †MEDIAN OVER RUNS THAT BALANCED.

		Nominal		Monte-Carlo		
Controller	Var.	t_s [s]	$\int \|\tau\|^2$	bal. [%]	95 % CI	IAE†
LQR	cl.	0.08	0.010	99.3	[96, 100]	5.7
LQR	MM	0.08	0.010	100.0	[97, 100]	5.8
MPC	cl.	0.08	0.015	96.7	[92, 99]	5.2
FBL+LQR	cl.	0.06	0.102	36.0	[28, 44]	0.3
FBL+LQR	MM	0.06	0.105	82.0	[75, 88]	3.4
SMC	cl.	0.06	0.100	36.0	[28, 44]	0.4
SMC	MM	0.06	0.100	94.0	[88, 97]	4.7
Backstepping	cl.	0.06	0.131	34.7	[27, 43]	0.3
Backstepping	MM	0.06	0.131	85.3	[78, 91]	3.8
PID	cl.	0.06	0.243	33.3	[26, 42]	0.5
PID	MM	0.06	0.242	96.7	[92, 99]	5.1

comes from a larger k_h ($0.16 \to 0.26$ for FBL+LQR), paid for in attitude error; exchanging k_h alone reproduces it, so managed survival is set mainly by k_h. Table I keeps the gains of the submitted study.

On paired plants (McNemar; hierarchical win ratio: survival, accuracy, time to fall) momentum management wins 1.8–2.0 pairs per loss on FBL+LQR, SMC, backstepping and PID ($p < 10^{-3}$, Holm over fifteen tests). Where both survive it is *less* accurate (rank-biserial $r > 0.85$): it pays for survival with a permanent tilt.

E. What the MPC baseline settles

With the submitted gains the MPC outperforms managed FBL+LQR and backstepping (22/0 and 17/0 discordant plants, McNemar $p < 10^{-4}$) and ties managed SMC and PID; after the re-tuning all are level. Momentum management buys no survival advantage over the MPC. We make no computational claim.

The MPC never reaches its momentum bound (peak 858 of $1000\,\mathrm{rad\,s^{-1}}$): like the LQR, it is consistent with surviving through the lean its weight on stored momentum induces; each of its five falls follows 16–18 solver fallbacks. What (14) adds is retrofit — a reference offset leaving the inner loop untouched — which takes a PID from 33.3 to 96.7 %, level with the MPC.

F. Limitations

Sweeping b_p over three decades leaves the unloaded momentum identical to three significant figures: the unloading is gravity, not pivot dissipation. Four limitations bound the conclusions. The pivot is an ideal spherical joint (A1); a real vertex adds stiction, which widens the equilibrium of Proposition 2 into $\theta^\star \pm \tau_s/(Mgl)$, adds a drilling torque that breaks Lemma 1, and can slip. The effect that does bite at this scale is a *constant* attitude-estimate bias b: since any equilibrium holds the true gravity-balance attitude (Proposition 2), (14) absorbs it as additional standing momentum, to first order $Mgl\|Pb\|/k_h$, while a design regulating the measured error to zero meets it as an offset. A bias of 0.64° is torque-equivalent to the whole $2\,\mathrm{mm}$ offset studied here, and IMU-to-diagonal alignment on this class of machine is of that order. We model neither an estimator nor such a bias. No design can unload a disturbance along $\boldsymbol{n}$. The recoveries need pivot friction up to $\mu \approx 0.35$; $\eta_{\lim}$ depends on the unidentified bearing friction (0.62 at $3b_w$); a 1.1 % parallel-axis inertia omission changes none of 120 paired outcomes. The nominal study is in-sample (a tuning scenario). And the study is entirely simulation-based.

VII. CONCLUSIONS

On a corner-balancing Cubli the wheels run out of speed before torque, and only the destabilizing gravity torque can unload them. Every equilibrium of the disturbed plant holds a gravity-balanced lean, momentum along $\boldsymbol{n}$ cannot be unloaded with an ideal pivot, and the sustainable speed is set by $\eta_{\lim}$, not by $\Omega_{\max}$. In simulation, designs that regulate the attitude error to zero fail under a millimetre-scale offset, while a gravity-torque unloading law, the momentum-weighted LQR and the MPC survive by holding a lean; the augmentation does not outperform the latter two. These rates are specific to the simulated plants, offsets and horizon; certified recovery regions, vertex friction and experiments remain open.

REPRODUCIBILITY

Plant, controllers, tuning and Monte-Carlo scripts, per-run results with their seeds, the supplementary tables and the scripts that regenerate every number, table and figure (e.g. `make_numbers.py`) are available at https://github.com/JoseSegundoM/cubli_momentum_budget.

REFERENCES

[1] M. W. Spong, P. Corke, and R. Lozano, "Nonlinear control of the reaction wheel pendulum," *Automatica*, vol. 37, no. 11, pp. 1845–1851, 2001.

[2] M. Gajamohan, M. Muehlebach, T. Widmer, and R. D'Andrea, "The Cubli: A reaction wheel based 3D inverted pendulum," in *2013 European Control Conference (ECC)*, 2013, pp. 268–274.

[3] M. Muehlebach and R. D'Andrea, "Nonlinear analysis and control of a reaction-wheel-based 3-D inverted pendulum," *IEEE Transactions on Control Systems Technology*, vol. 25, no. 1, pp. 235–246, 2017.

[4] U. Wasiwitono, D. K. Arif, L. Nurahmi, and B. Pramujati, "Modeling and control of reaction wheel pendulum with feedback linearization and LQR control," in *Smart Innovation in Mechanical Engineering*, ser. Lecture Notes in Mechanical Engineering. Springer Nature Singapore, 2025, pp. 315–322.

[5] H. Huang, Z. Li, Z. Guo, and J. Guo, "Prescribed performance adaptive balance control for reaction wheel-based inverted pendulum-type Cubli rovers in asteroid," *Aerospace*, vol. 9, no. 11, p. 728, 2022.

[6] M. J. Sidi, *Spacecraft Dynamics and Control: A Practical Engineering Approach*. Cambridge University Press, 1997.

[7] B. K. Powell, "Gravity-gradient momentum management," *Journal of Spacecraft and Rockets*, vol. 9, no. 6, pp. 385–386, 1972.

[8] M. A. Shahraki and L. Lessard, "Spacecraft attitude control under reaction wheel constraints using control Lyapunov and control barrier functions," in *2025 American Control Conference (ACC)*, 2025, pp. 947–952.

[9] B. Vaidya, M. Shomin, R. Hollis, and G. Kantor, "Operation of the ballbot on slopes and with center-of-mass offsets," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2015, pp. 2383–2388.

[10] W.-H. Huang, P. T.-T. Nguyen, D.-D. Nguyen, H.-P. Doan, M.-Y. Chuang, and C.-H. Kuo, "Stability control and path tracking of a self-balancing bicycle with a reaction wheel," *IEEE/ASME Transactions on Mechatronics*, vol. 30, no. 4, pp. 2928–2936, 2025.

[11] E. Garone, S. Di Cairano, and I. Kolmanovsky, "Reference and command governors for systems with constraints: A survey on theory and applications," *Automatica*, vol. 75, pp. 306–328, 2017.

[12] F. Markley, R. Reynolds, F. Liu, and K. Lebsock, "Maximum torque and momentum envelopes for reaction wheel arrays," in *Proc. AIAA Guidance, Navigation, and Control Conf.*, 2009.